Ci6X Supplemental Document for XDS 4

This supplement is specific to the instrument listed above and should be used in conjunction with the X-Rite Driver System v4 (XDS 4) Technical Specification.

Instrument Commands

AutoName Clear

---

autoname clear {arg1}

If arg1 is not provided, all prefixes are deleted (erased and no longer available). If arg1 is provided then the active prefix is disabled and must be re-selected if it is to be used again (See the *AUTONAME SELECT* command).

Arguments

- *arg1*
  - SAMPLE – Disables the active sample prefix.
  - STANDARD – Disable the active standard prefix.

Response

Returns command/status only. No data.

Example

Disable the active sample prefix.

AUTONAME CLEAR sample

Remove all sample and standard prefixes.

AUTONAME CLEAR
<00>

AutoName Get

autoname get arg1 {arg2}

Used to retrieve the current active prefix or all stored prefixes.

Arguments

- *arg1*
  - Sample – returns all stored prefixes that can be used to auto-name samples
  - Standard – returns all stored prefixes that can be used to auto-name standards
  - Active – must specify either sample or standard to retrieve the active prefix
- *arg2 – must be sample or standard*

Response

Returns one or more ASCII or UTF-16 prefixes.

Example

Get all available sample prefixes

AUTONAME GET SAMPLE
SampPre1 SampPre2 SampPre3 SomePrefix

Get all available standard prefixes

AUTONAME GET STANDARD
OnePrefix AnotherOne SomePrefix

Get the active sample prefix

AUTONAME GET ACTIVE SAMPLE
AnotherOne

AutoName Select

autoname select arg1 arg2

Used to select the prefix that will be used to auto-name samples or standards.

Arguments

- *arg1*
  - Sample – next argument will be a sample prefix
  - Standard – next argument will be a standard prefix
- *arg2*
  - must be a prefix that is already loaded onto the instrument

Response

Returns command status only. No Data

Example

Select a sample prefix

AUTONAME SELECT SAMPLE SomePre
<00>

AutoName Set

---

autoname set arg1 arg2

Sets the prefix to be used to auto-name samples and/or standards. If no auto-name prefix is available or this feature is disabled, the sample or standard is stored with a time stamp in place of the name. By default, the auto-name feature for both samples and standards is disabled. To enable sample auto-name choose the desired setting for the AUTONAMESAMPLE parameter, send the following command: Param Set AutoNameSample 1 to turn on and Param Set AutoNameSample 0 to turn off.

Up to 32 prefixes may be stored. After all prefixes have been loaded, an *AUTONAME STORE* command must be issued to store data. The active prefix is set using the *AUTONAME SELECT* command.

Arguments

- *arg1*
  - Sample – next argument supplied will be a prefix for auto-naming samples
  - Standard – next argument supplied will be a prefix for auto-naming standards
- *arg2*
  - specify the name for the prefix. Can be 16 ASCII or UTF-16 characters

Response

Returns command status only. No Data

Example

Set the sample prefix to Sample

AUTONAME SET SAMPLE Sample
<00>

Set the standard prefix to Standard

AUTONAME SET STANDARD Standard
<00>

AutoName Store

autoname store

Stores all loaded prefixes.

Arguments

None.

Response

Returns command status only. No Data

Example

Store the loaded prefixes

AUTONAME STORE
<00>

Bluetooth Config

bluetooth config subcmd

Configures the bluetooth module and supports low-level interaction with the module.

Arguments

- *subcmd* – what action to take with the bluetooth device
  - *AlwaysOn* – issue this first if you are currently talking over USB but want to configure bluetooth

Response

Returns command/status only

Example

Set bluetooth radio always on

BLUETOOTH CONFIG ALWAYSON
<00>

Bluetooth Get

bluetooth get subcmd

Retrieves information about the bluetooth module.

Arguments

- *subcmd* – what information to retreive
  - *Addr* – retrieves the bluetooth modules radio address

Response

Returns the bluetooth radios radio address as 12 alphanumeric characters.

Example

Retrieves the instruments bluetooth radio address

BLUETOOTH GET ADDR
0012AC25BF34

Bluetooth Reset

bluetooth reset

Forces a hard reset of the module

Arguments

None.

Response

Returns command/status only

Examples

Resets The Module

BLUETOOTH RESET
<00>

Buttons Disable/Enable

buttons disable
buttons enable

Enables or Disables the button interface. This is used by software to lock out button presses during certain operations to keep the instrument in a known state. When the buttons are disabled, only the power off button remains operational (must be held for approximately three seconds).

Arguments

none

Response

Only Command Status/Error returned

Example

Disable user button presses.

BUTTONS DISABLE
<00>

CalPlaque Get

calplaque get option

Causes the instrument to return white calibration values for the specified geometry.

Arguments

- *option*
  - *geometry – The geometry to retrieve reflectances or other trait. It is one of the following:*
    - spin
    - spex
  - *serial – The serial number of the calplaque.*

Response

Returns 33 floating point reflectance values corresponding to the calibration plaque for the specified geometry. There is a value for each wavelength from 390nm to 710nm at 10nm increments. If serial was specified, a single ASCII text string is returned containing the serial number

Examples

Fetch the calibration plaque values for white, specular included

CALPLAQUE GET spin
90.12 90.34 91.56 …(x33)

Fetch the calplaque serial number

CALPLAQUE GET serial
P000001

CalStatus Get

calstatus get

Arguments

none

Response

This command returns the next required calibration step. This step will be text (BLACK, WHITE, EXPIRED) if a normal calibration problem. If it returns an error code, it is likely to require service. If all steps are completed, it returns OK, followed by the temperature in degrees C and the number of minutes until the *WHITE* calibration expires.

If a calibration is expired, the text *EXPIRED* followed by which calibration is expired will be returned. If a calibration is expired, perform the necessary steps to re-calibrate the instrument.

Example

A black calibration is required

CALSTATUS GET
BLACK

Calibration is ok and will expire in 4 hours (240 minutes)

CALSTATUS GET
OK 25.0 240

CalWhite Abort

calwhite abort

This command is used to abort a white calibration prior to a calibration being started.

Arguments

none.

Response

Returns command/status only.

Example(s)

abort a white calibration.

CALWHITE ABORT
<00>

Captouch Get

captouch get option

Returns the specified information from the CapTouch board.

Arguments

- *option – This is the item to get.*
  - *version – Returns the version string of the Captouch firmware.*
  - *status – Returns the key press status.*
    - The *Position* is a value from 0 to 255 indicating the position being touched on the key circle.
      - DOWN is approximately 0 (or 255).
      - LEFT is approximately 64.
      - UP is approximately 128.
      - RIGHT is approximately 192.
    - The *CenterStatus* is one of the following:
      - 0 = no key pressed.
      - 1 = a non-center key is pressed (UP, DOWN, LEFT, RIGHT).
      - 2 = the center key (ENTER) is pressed.
      - 3 = invalid key entry (full hand over the keypad).
    - The *Touched* status is 0 if no key pressed or 1 if any key is pressed.

Response

The status which is followed by the requested data.

Examples

Return the status of the Captouch board when the DOWN key is pressed.

CAPTOUCH get status
Position: 254
CenterStatus: 1
Touched: 1

Return the version of the Captouch board.

CAPTOUCH get version
X-Rite CapTouch 1.887

CheckPlaque Get

checkplaque get subcmd

Causes the instrument to return green check plaque serial number or values for the specified geometry (31 reflectance points).

Arguments

- *geometry – The geometry to retrieve reflectance data for. It is one of the following:*
  - *spin*
  - *spex*
- *serial – retrieve the serial number of the green check plaque*

Response

Returns 31 floating point reflectance values corresponding to the check plaque. There is a value for each wavelength from 400nm to 700nm at 10nm increments. If serial, the serial number is returned as ASCII text.

Examples

Fetch the check plaque values for specular included

CHECKPLAQUE GET SPIN
79.03 45.23 32.31 …(x31)

Fetch the check plaque serial number

CHECKPLAQUE GET SERIAL
G000123

Clock Get

clock get

Returns the date and time from the realtime clock.

Arguments

none

Response

The status is followed by the day, month, date, time, year in standard posix format.

Example

Return the date time of the clock in the instrument.

CLOCK GET
Tue Feb 9 11:39:06 2010

Clock Set

clock set month date year hour min sec

Set the realtime clock. All unspecified values are set to defaults. **NOTE** the earliest date setable by the system is Jan 1 2000 at midnight.

Arguments

- *month – as a number between 1 and 12*
- *date – as a number between 1 and 31*
- *year – as the last two digits of the year.*
- *hour – as a the 24 hour format 0 – 23*
- *min – as the number of minutes into the hour 0 – 59*
- *sec – as the number of seconds into the minute 0 – 59*

Response

Just Status.

Example

This will set the clock to February 24, 2010 at 3:45:00 PM

CLOCK SET 2 24 10 15 45
<00>

Coeff Get

coeff get arg1

This command is used to retrieve the default coefficients used in the dE2000, dE94, and dEcmc calculations.

Arguments

- *arg1*
  - *Specify which dE to extract coefficients for. Must be one of dE2000, dE94, or dEcmc.*

Response

Will return the default values used for specified dE calculation. Will be either 2 or 3 floating point values.

Examples

Get the dE2000 coefficients.

COEFF GET DE2000
5.0 1.25 3.25

Get the dECMC coefficients.

COEFF GET DECMC
3.5 1.25

Coeff Set

---

coeff set arg1 c1 c2 {c3}

This command is used to set the default coefficients used in the dE2000, dE94, and dEcmc calculations. **You must issue a <u>Coeff Store</u> command to save these settings**. The factory defaults are as follows:

- *dE94*
  - *kL – 2.00*
  - *kC – 1.00*
  - *kH – 1.00*
- *dE2000*
  - *kL – 2.00*
  - *kC – 1.00*
  - *kH – 1.00*

- *dEcmc*
  - *l – 2.00*
  - *c – 1.00*

Arguments

- *arg1*
  - *Specify the dE the coefficients are for. Must be one of dE2000, dE94, or dEcmc*
- *c1*
  - *The first coefficient (kL or l). Float value, req'd for all dE arguments.*
- *c2*
  - *The second coefficient (kC or c). Float value, req'd for all dE arguments.*
- *c3*
  - *The third coefficient (kH). Float value, only req'd for dE2000 and dE94.*

Response

Command status only.

Examples

Set the dE2000 coefficients.

COEFF SET DE2000 5.0 1.25 3.25
<00>

Set the dEcmc coefficients.

COEFF SET DECMC 3.5 1.9
<00>

Coeff Store

---

coeff store

Saves any coefficients specified using Coeff Set command.

Arguments

None.

Response

Command Status Only

Examples

Store the coefficients

COEFF STORE
<00>

DefaultTol Set

---

defaulttol set opts args

Sets the default tolerance to be used in the instrument. Note that the value(s) are not saved until you issue a store command.

Arguments

- *opts – specify which option you want to set. One of the following:*
  - *colorspace – the default colorspace*
  - *illumobs – the default illuminant/observer*
  - *geometry – the default geometry*
  - *tolerance – the default tolerance values (+/-)*
- *args – the value(s) to set the specified option to*
  - *when **opts** is colorspace: single argument, the desired colorspace string, list is available delta colorspaces:*

    - *DLAB* – L*a*b* tolerance type.
    - *DLCH* – L*C*H* tolerance type.
    - *DXYZ* – XYZ tolerance type.
    - *DYXY* – Yxy tolerance type.
    - *DLUV* – L*u*v* tolerance type.
    - *DHUNTERLAB* – Hunter Lab tolerance type.
    - *DELAB* – Delta E tolerance type.
    - *DECMC* – Delta Ecmc tolerance type.
    - *DE2000* – Delta E 2000 tolerance type.
    - *DE94* – Delta E 94 tolerance type.
    - *DYI98* – E313-98 yellowness index tolerance type.

- *DYI73* – E313-73 yellowness index tolerance type.
- *DYI25* – D1925 yellowness index tolerance type.
- *DWI98* – E313-98 whiteness index tolerance type.
- *DWI73* – E313-73 whiteness index tolerance type.
- *DWIBERGER* – Berger whiteness index tolerance type.
- *DWIHUNTER* – Hunter whiteness index tolerance type.
- *DWISTENSBY* – Stensby whiteness index tolerance type.
- *DWITAUBE* – Taube whiteness index tolerance type.

- *when **opts** is illumobs: single argument, the desired illum/obs string, list is available illum/obs:*

  - *A/2*
  - *A/10*
  - *C/2*
  - *C/10*
  - *D50/2*
  - *D50/10*
  - *D65/2*
  - *D65/10*
  - *F2/2*
  - *F2/10*
  - *F7/2*
  - *F7/10*
  - *F11/2*
  - *F11/10*
  - *F12/2*
  - *F12/10*

- *when **opts** is geometry : single argument, the desired geometry string, one of spin, spex, or spin/spex*
- *when **opts** is tolerance : six arguments, the desired delta values, 3 pairs of values, order is -, +, -, +, -, +*

Response

- Returns command status only.

Examples

Sets the default tolerance colorspace to delta LCh

DEFAULTTOL SET COLORSPACE DLCH
<00>

Sets the default tolerance illum/obs to D65/10

DEFAULTTOL SET ILLUMOBS D65/10
<00>

Sets the default tolerance geometry to spex

DEFAULTTOL SET GEOMETRY SPEX
<00>

Sets the default tolerance plus/minus values

DEFAULTTOL SET TOLERANCE -0.83 1.35 -1.00 1.25 -0.98 1.35
<00>

DefaultTol Store

defaulttol store

Saves the values in the scratch space to the default tolerance. Assumes you have used the *set* command to set the desired values.

Arguments

None

Response

- Command status only

Examples

Stores the default tolerance values

DEFAULTTOL STORE
<00>

Display Refresh

display refresh {option}

Overview

This command is used to enable or disable the display refresh capability. It also locks out the keys so a user can't change anything. The exception to the key lockout are the measure switch/button and the power button. The purpose of this command is to inhibit time consuming display refreshes and lock out any manual user interface when a lot of USB command activity is required. The "*display refresh disable*" locks out the user interface. The "*display refresh enable*" restores normal operation.

Arguments

- *option*
  - *DISABLE – Locks out the user interface buttons except the measure switch/button and power button and inhibits display updates.*
  - *ENABLE – Enables the user interface buttons and display updates.*
  - *If no option is provided the display is forced to update and the user interface buttons are re-enabled.*

Response

DISPLAY Status

Examples

Disable buttons and display refreshing.

DISPLAY REFRESH DISABLE
<00>

Enable buttons and display refreshing.

DISPLAY REFRESH ENABLE
<00>

Refresh the display.

DISPLAY REFRESH
<00>

ErrorLog Clear

errorlog clear

Clears the contents of the error log.

Arguments

none

Response

Returns command/status only. No data.

Example

Clear all the entries in the error log

ERRORLOG CLEAR
<00>

ErrorLog Get

errorlog get

Throughout the operation of the instrument, situations may arise that generate warnings, errors, or alternate status codes. These codes are recorded in the errorlog for future reference and remain there until the errorlog clear command is used. A list of these codes and their descriptions may be found in the Status Codes section. ErrorLog Get causes the instrument to transmit the list of error/status codes recorded in the error log.

Arguments

none

Response

Returns the list of error/status codes that were generated since the last time the log was cleared. The list varies in length depending on how many different error/status conditions have occurred. The first argument of the response provides the number of different codes that exist in the list. This is followed by pairs of arguments where the first item is the error code and the second is the number of times that error occurred. Line feeds are inserted before each pair for easier readability.

Example

Fetch the error log listing

ERRORLOG GET
4
0×01 3
0×32 1
0×03 2
0×02 1

File Summarize

file summarize

Fetches a summary of the file system usage. Instead of breaking it down by volume, this command tries to summarize in terms of usage that an end-user might care about.

Arguments

None.

Response

Status Code followed by the following:

- *Bytes used and total bytes of the file system*
- *Number used and number of total samples*
- *Number used and number of total standards*
- *Number used and number of total projects*
- *Number of images on the instrument*

Return the information for the Flash File System

FILE SUMMARIZE
system: 415736 8257536
samples: 46 4000
standards: 14 1000
projects: 5 100
jobs: 2 10
images: 3

GUI Get

gui get

This command returns the title of the current GUI screen being shown on the instrument.

Arguments

None.

Response

Responds with the current *screen* being displayed.

- *Screen Titles:*
  - MAIN
  - MEASURE
  - CALIBRATE
  - CONFIG

Example(s)

get the current screen being displayed

GUI GET
MAIN

GUI Set

gui set screen

This command forces the the instrument to display a specified GUI *screen*.

Arguments

- *screen*
  - main – *force the gui to the main screen*
  - measure – *force the gui to the measure screen*
  - calibrate – *force the gui to the calibrate screen*
  - config – *force the gui to the config screen*

Response

Command Status Only.

Example(s)

set the gui to the measure screen

GUI SET MEASURE
<00>

Meas Get

meas get geometry

Returns the reflectance results of the last measurement for the specified *Geometry*. The error code will be returned instead if the last measurement failed or no measurement has yet been performed.

Arguments

- *geometry*
  - *SPIN – returns specular included reflectance data.*
  - *SPEX – returns specular excluded reflectance data.*

Response

This command returns 31 floating point values for the geometry as specified. There is a value for each wavelength from 400nm to 700nm at 10nm increments. If UseExtendedRefls is enabled through Params, then 40 floating point values are returned instead. The format for 40 point data is 400nm repeated 4 times, then 400nm through 700nm, then 700nm repeated 5 times.

Example

Fetch specular included reflectance values for the last measurement

MEAS GET SPIN
90.3412 91.5613…(SPIN values)

Meas Run

meas run

The instrument will perform a measurement if the read switch (shoe switch) or the read button is activated. The meas run command can be used to perform a measurement remotely. The results from the measurement can be obtained using the meas get command. The instrument does not automatically transmit the results.

This command also uses the meas done? command to determine the status of the operation. None Yet will be returned until the operation starts, after which you will get Pending, OK, or error codes as usual.

Arguments

The default operation of the meas command is to perform a measurement.

Response

- Returns the status only.

Example

Shows the measurement is complete with new data, and shows what happens if you issue another done command w/o doing another measurement

meas run
MEAS 0×05 1000 3000
meas done?
MEAS 0×00
meas done?
MEAS 0×07

ModelConfig Add

modelconfig add opt arg

This command adds items to the current configuration settings. A *modelconfig store* command must be issued after you have issued all your add commands to save the files.

Arguments

- *opt – The setting to add to. One of:*
  - *allowedcolorspaces*
  - *allowedillumobs*
- *arg – the item(s) to add. Can be a single item or a list of items.*

- *Colorspace options for allowedcolorspaces*

  - **REFL** – 31 reflectance values
  - **LAB** – L*a*b* colorspace
  - **LCH** – L*C*h colorspace
  - **XYZ** – XYZ colorspace
  - **YXY** – Yxy colorspace
  - **LUV** – Luv colorspace
  - **HUNTERLAB** – Hunter Lab colorspace
  - **GRAYSCALE** – gray scale
  - **MUNSELL** – Munsell Notation
  - **GLOSS** – gloss value for the measurement specified in *index*.
  - **MI** – Metamerism Index
  - **MI6172** – D6172 Metamerism Index
  - **YI98** – E313-98 yellowness index
  - **YI73** – E313-73 yellowness index
  - **YI25** – D1925 yellowness index
  - **WI98** – E313-98 whiteness index
  - **WI73** – E313-73 whiteness index
  - **WIBERGER** – Berger whiteness index
  - **WIHUNTER** – Hunter whiteness index
  - **WISTENSBY** – Stensby whiteness index
  - **WITAUBE** – Taube whiteness index
  - **PASS/FAIL –** Pass or fail indicator based on tolerance from a standard

- *Illumimant/Observer options for allowedillumobs*

  - A/2
  - A/10
  - C/2
  - C/10
  - D50/2
  - D50/10
  - D65/2
  - D65/10
  - F2/2
  - F2/10
  - F7/2
  - F7/10

- F11/2
- F11/10
- F12/2
- F12/10

Response

- Command status only

Examples

Add two new illum/obs to the current list

MODELCONFIG ADD ALLOWEDILLUMOBS F2/2 F2/10
<00>

ModelConfig Delete

---

modelconfig delete opt args

This command deletes items off of the current model configuration settings for the specfied *opt*. A modelconfig store command must be issued after you have issued all your delete commands to save the files.

Arguments

- *opt – One of:*
  - *allowedcolorspaces*
  - *allowedillumobs*
- *args – list of items to be removed from the specified setting*

Response

- Returns status only.

Examples

Delete 3 colorspaces from the allowed colorspace list

MODELCONFIG DELETE ALLOWEDCOLORSPACES XYZ LUV YI
<00>

ModelConfig Get

modelconfig get opt

This command gets the current model config for the requested setting.

Arguments

- *opt – One of:*
  - *allowedcolorspaces*
  - *allowedillumobs*

Response

- Returns the current model configuration for the given option.

Examples

Get the allowed colorspaces

MODELCONFIG GET ALLOWEDCOLORSPACES
LAB LCH

ModelConfig Set

modelconfig set opt args

This command sets up certain model specific configuration settings. A modelconfig store command must be issued after you have issued all your set commands to save the files.

Arguments

- *opt – One of:*
  - *allowedcolorspaces – this sets up which colorspaces are selectable/viewable from the gui.*
  - *allowedillumobs – this sets up which illum/observers are selectable/viewable from the gui.*
- *args – a list of strings that set up the model specific configs for the desired setting.*
  - *Colorspace options for allowedcolorspaces*
    - **REFL** – 31 reflectance values
    - **LAB** – L*a*b* colorspace
    - **LCH** – L*C*h colorspace

- **XYZ** – XYZ colorspace
- **YXY** – Yxy colorspace
- **LUV** – Luv colorspace
- **HUNTERLAB** – Hunter Lab colorspace
- **GRAYSCALE** – gray scale
- **MUNSELL** – Munsell Notation
- **GLOSS** – gloss value for the measurement specified in *index*.
- **MI** – Metamerism Index
- **MI6172** – D6172 Metamerism Index
- **YI98** – E313-98 yellowness index
- **YI73** – E313-73 yellowness index
- **YI25** – D1925 yellowness index
- **WI98** – E313-98 whiteness index
- **WI73** – E313-73 whiteness index
- **WIBERGER** – Berger whiteness index
- **WIHUNTER** – Hunter whiteness index
- **WISTENSBY** – Stensby whiteness index
- **WITAUBE** – Taube whiteness index
- **PASS/FAIL** – Pass or fail indicator based on tolerance from a standard

- *Illumimant/Observer options for allowedillumobs:*

  - **A/2**
  - **A/10**
  - **C/2**
  - **C/10**
  - **D50/2**
  - **D50/10**
  - **D65/2**
  - **D65/10**
  - **F2/2**
  - **F2/10**
  - **F7/2**
  - **F7/10**
  - **F11/2**
  - **F11/10**
  - **F12/2**
  - **F12/10**

Response

- Command status only

Examples

Set the allowed colorspaces to LAB, Lch, XYZ

MODELCONFIG SET ALLOWEDCOLORSPACES LAB LCH XYZ
<00>

ModelConfig Store

modelconfig store

This command saves any model configuration settings that have been set using the modelconfig set command. Once this has been done, any new modelconfig set commands will start this process over, replacing the old values. Therefore it is important that you keep track of which parameters you actually want to override and resend them if you want to add or modify another.

Arguments

None.

Response

- Command status only

Examples

Store the currently configured model configuration settings

MODELCONFIG SET ALLOWEDCOLORSPACES LAB LCH LUV
<00>
MODELCONFIG STORE
<00>

Opacity Get

opacity get

Returns the opacity coefficients.

Arguments

none

Response

The status is followed by four opacity coefficients in floating point format.

Example

Return the opacity coefficients in the instrument.

OPACITY GET
0.0123 0.0456 0.0567 0.1012

Opacity Set

opacity set k1 k2 k3 k4

This command is used to set the coefficients used in the opacity calculation.

Arguments

k1 – k4 are the four opacity coefficients to be set (floating point format).

Response

Command status only.

Examples

Set the opacity coefficients.

OPACITY SET 0.0123 0.0456 0.0567 0.1012
<00>

Opacity Store

opacity store

Used to store the opacity coefficient data that was previously loaded using the *opacity set* command.

Arguments

none

Response

Returns command/status only. No data.

Example

Store the present opacity coefficient settings.

OPACITY STORE
<00>

Param Default

param default param_name

Sets a parameter back to default values. You may optionally specify 'all' as the parameter name to set all of them to defaults.

Arguments

- *param_name – The name of the parameter to set back to the defaults. Multiple parameters can be listed if desired, separated by spaces.*
- *all – instead of resetting an individual parameter, this will set all parameters at the current access level to defaults.*

Response

Returns command/status only. No data.

Examples

Set all parameters to defaults

PARAM DEFAULT all
<00>

Set the Cal White Timeout and Volume Level parameters to defaults

PARAM DEFAULT CalWhiteTimeout VolumeLevel
<00>

Param Get

param get param_name

Param Get is used to check instrument configuration parameters. This is the primary method of customizing the user experience.

Arguments

- *param_name – the name of the parameter to retrieve.*

Response

Returns a 32-bit number as text which is either the user-configured value for that parameter, or the default if it hasn't been specified.

Example

Get the value for White Cal Hours

PARAM GET CalWhiteTimeout
8

Param Set

param set param_name value

Param Set is used to set instrument configuration parameters. This is the primary method of customizing the user experience. Params are specified with min and max values, so setting a parameter will automatically generate and error status code if out of range. They are treated as unsigned integers unless special handling has been added.

Arguments

- *param_name – the name of the parameter to retrieve.*
- *value – the value to set the parameter to. This is a single integer value.*

Response

Returns command/status only. No data.

Example

Set the White Cal Timeout to 8 hours

PARAM SET CalWhiteTimeout 8
<00>

Project Clear

project clear name

Clears all data associated with the project specified in the *name* argument and removes the project from the instrument.

Arguments

- *name*
  - *specifies the name of the project to be removed. If the name is "all" then all projects are removed.*

Response

Returns command/status only. No data.

Example

Remove a project

PROJECT CLEAR MyProject
<00>

Remove all projects

PROJECT CLEAR ALL
<00>

Project Get

project get arg

Retrieves information about the active project.

Arguments

- *arg*
  - *Name – returns the name of the active project. Optionally a number may be specified to return the name of the project in that slot*
  - *Names – returns a list of all the projects currently on the instrument*
  - *StandardName – must specify the project relative slot number of the standard. 0 through max number of standards in projects – 1*
  - *StandardNames – returns a list off all the standards in the active project. if more standards are present than what can fit in a message a '… ###' will be returned with the message. The ### will be the id of the next standard. Issue this command again with a numerical argument to start at that standard.*
  - *NumStandards – returns the number of standards in the active project*
  - *Sample – returns sample information with all the sample options as the* SAMPLE GET *command except the index is project based. (See also* SAMPLE GET*).*
  - *Config – returns the ProjectConfigMask for the active project*

Response

Varies based on arguments.

Example

Get the name of the active project

PROJECT GET NAME
MyProject\u8878\u9989

Get the name of the 3rd project

PROJECT GET NAME 3
ThirdOne\u8878

Get the name of all projects on the instrument

PROJECT GET NAMES
MyProject AnotherOne ThirdOne\u8878

Get the name of the 3rd standard in this project

PROJECT GET STANDARDNAME 2
TheThirdStandard

Get the name of all standards in the active project

PROJECT GET STANDARDNAMES
First\u8768\u4345 AnotherStandard TheThirdStandard \u4434\u9987\u3485\u8675

Get the name of all standards in the active project multiple packets

PROJECT GET STANDARDNAMES
First\u8768\u4345 AnotherStandard TheThirdStandard \u4434\u9987\u3485\u8675 … 4
PROJECT GET STANDARDNAMES 4
AFifthStandard TheSixthOne LastStandard

Get the number of samples stored in this project.

PROJECT GET SAMPLE COUNT
14

*Get the L*a*b* data (specular included) for sample 8 in this project.*

PROJECT GET SAMPLE LAB 8 SPIN
36.28 33.58 16.02

Project New

project new name

Creates a new project on the instrument.

Arguments

- *name*
  - *specifies the desired name for the new project. Can be up to 32 ASCII or UTF-16 characters.*

Response

Returns the id of the newly created project

Example

Create a new project

PROJECT NEW MyProject
1

Create a new project

PROJECT NEW My\u4568\u3423Project
2

Project Select

project select name

Makes *name* project the new-active project.

Arguments

- *name*
  - *specifies the name of the project to be made active.*

Response

Returns command/status only. No data.

Example

Select a new project

PROJECT SELECT MyProject
<00>

Project Set

project set arg1 arg2

Used to set the active projects configuration mask (ProjectConfigMask). The configuration mask is used to enable/disable users from changing the project.

Arguments

- *arg1*
  - *Config – next argument is the configuration mask*
    - arg2 – a unsigned integer representing the project configuration mask for the active project

Response

Command Status Only.

Example

Set the active project configuration mask to disable renaming the project and adding standards

PROJECT SET CONFIG 0×10000046
<00>

Project Store

project store

Saves the active project. Mostly used after you add/remove/edit any standards associated with the active project.

Arguments

none.

Response

Returns command/status only. No data.

Example

Store the current project

PROJECT STORE
<00>

Sample Clear

sample clear option

Used to delete one or more sample measurements stored in the instrument.

Arguments

- *option*

- *ALL – Deletes the entire sample database.*
- *LAST – Deletes the last stored sample.*
- *PROJECT – Deletes all samples stored in the active project.*
- *index – A number indicating which stored sample is to be deleted (zero indicates the last stored sample).*

Response

Returns command/status only. No data.

Example

Clear all the stored samples.

SAMPLE CLEAR ALL
<00>

Clear the last sample only.

SAMPLE CLEAR LAST
<00>

Alternate method to clear the last sample only.

SAMPLE CLEAR 0
<00>

Clear sample number 6.

SAMPLE CLEAR 6
<00>

Clear all samples in the active project.

SAMPLE CLEAR PROJECT
<00>

Sample Get

sample get option {index} {geometry}

Returns information regarding a measurement. *option* specifies the type of information the instrument is to provide. The *index* is the number of a stored measurement. An index of 0 indicates

the last measurement. The *geometry* parameter is either SPIN or SPEX and is required for all options unless indicated otherwise.

For options that return the difference between the measurement and a standard, the standard used is the one active at the time of this command (or the closest one if auto standard is active), not necessarily the standard at the time of the measurement.

Arguments

- *option*
  - *REFL – returns 31 reflectance values for the measurement specified in index.*
  - *LAB – returns L*a*b* values for the measurement specified in index.*
  - *LCH – returns L*C*h values for the measurement specified in index.*
  - *XYZ – returns XYZ values for the measurement specified in index.*
  - *YXY – returns Yxy values for the measurement specified in index.*
  - *LUV – returns Luv values for the measurement specified in index.*
  - *HUNTERLAB – returns Hunter Lab values for the measurement specified in index.*
  - *GRAYSCALE – returns gray scale values (Stain and Color) for the measurement specified in index.*
  - *MUNSELL – returns Munsell Notation values for the measurement specified in index.*
  - *GLOSS – returns gloss value for the measurement specified in index.*
  - *MI – returns Metamerism Index value for the measurement specified in index.*
  - *MI6172 – returns D6172 Metamerism Index value for the measurement specified in index.*
  - *YI98 – returns E313-98 yellowness index values for the measurement specified in index.*
  - *YI73 – returns E313-73 yellowness index values for the measurement specified in index.*
  - *YI25 – returns D1925 yellowness index values for the measurement specified in index.*
  - *WI98 – returns E313-98 whiteness index values for the measurement specified in index.*
  - *WI73 – returns E313-73 whiteness index values for the measurement specified in index.*
  - *WIBERGER – returns Berger whiteness index values for the measurement specified in index.*
  - *WIHUNTER – returns Hunter whiteness index values for the measurement specified in index.*
  - *WISTENSBY – returns Stensby whiteness index values for the measurement specified in index.*
  - *WITAUBE – returns Taube whiteness index values for the measurement specified in index.*
  - *DLAB – returns the difference between the measurement and the standard L*a*b* values for the measurement specified in index.*
  - *DLCH – returns the difference between the measurement and the standard L*C*h values for the measurement specified in index.*
  - *DXYZ – returns the difference between the measurement and the standard XYZ values for the measurement specified in index.*
  - *DYXY – returns the difference between the measurement and the standard Yxy values for the measurement specified in index.*

- *DLUV – returns the difference between the measurement and the standard Luv values for the measurement specified in index.*
- *DELAB – returns the difference value (E) between the measurement and the standard for the measurement specified in index.*
- *DECMC – returns the difference value (Ecmc) between the measurement and the standard for the measurement specified in index.*
- *DE2000 – returns the difference value (E2000) between the measurement and the standard for the measurement specified in index.*
- *DE94 – returns the difference value (E94) between the measurement and the standard for the measurement specified in index.*
- *DYI98 – returns the difference between the measurement and the standard E313-98 Yellowness Index values for the measurement specified in index.*
- *DYI73 – returns the difference between the measurement and the standard E313-73 Yellowness Index values for the measurement specified in index.*
- *DYI25 – returns the difference between the measurement and the standard D1925 Yellowness Index values for the measurement specified in index.*
- *DWI98 – returns the difference between the measurement and the standard E313-98 Whiteness Index values for the measurement specified in index.*
- *DWI73 – returns the difference between the measurement and the standard E313-73 Whiteness Index values for the measurement specified in index.*
- *DWIBERGER – returns the difference between the measurement and the standard Berger Whiteness Index values for the measurement specified in index.*
- *DWIHUNTER – returns the difference between the measurement and the standard Hunter Whiteness Index values for the measurement specified in index.*
- *DWISTENSBY – returns the difference between the measurement and the standard Stensby Whiteness Index values for the measurement specified in index.*
- *DWITAUBE – returns the difference between the measurement and the standard Taube Whiteness Index values for the measurement specified in index.*
- *SHADESORT – returns the 555 shade sort value. Valid only if shade sort is enabled and measurement occurred during a valid shade sort mode.*
- *VERBALDIFF – returns the difference verbally. Valid only if verbal difference is enabled and measurement occurred during a valid difference mode.*
- *OPACITY – returns the opacity value. Only valid after the last reading of the procedure.*
- *OPACITYTYPE – returns the opacity type value (0 = over black, 1 = over white, 2 = color at 100%).*
- *OPACITYILLUMOBS – returns opacity illuminant/observer value (see the illumobs parameter for list).*
- *COUNT – returns the number of measurements stored in the instrument. No other parameters required.*
- *NAME – returns the 32 character (codepoint) name of the sample.*

- *TIMESTAMP – returns the time (hh:mm:ss MM/DD/YY) the measurement was taken for the measurement specified in index.*
- *TEMP or TEMPERATURE – returns the temperature (degrees C) of the instrument at the time the measurement was taken for the measurement specified in index.*
- *APERTURE – returns the aperture of the instrument (SMALL, MEDIUM, or LARGE).*
- *AVERAGE – returns the number of readings averaged to create the measurement.*
- *PROJECTID – returns the project identification number associated with this sample.*
- *STANDARD – returns number of the closest standard to the measurement color.*
- *TAG1 – returns the first identification number tag.*
- *TAG2 – returns the second identification number tag.*
- *TAG3 – returns the third identification number tag.*
- *TAG4 – returns the fourth identification number tag.*
- *TAG5 – returns the fifth identification number tag.*
- *PASS/FAIL – returns the pass/fail status (IGNORED, PASS, WARNING, FAIL).*
- *CALSTATUS – returns the calibration status, temperature, and time remaining of the calibration at the time of the measurement.*

  - status may be: OK, BLACK, WHITE, or EXPIRED.
  - The temperature is in degrees Celsius.
  - The time remaining is the number of minutes until the calibration will run out.

- *TRANSFORMID – returns the identification number of the active transform for the measurement specified in index.*

  - Values 0 through 6 indicate the selected transform.
  - A value of 7 indicates no transform is active.
  - A value of 8 indicates the default transform is to be used. This occurs if no transform has ever been selected. The default will be for transforms to be turned off unless the *TRANSFORM SET DEFAULT* command has been used to change the default transform.

- *TRANSFORMGLOSS – returns the gloss correction value used by the transform calculations.*
- *TRIGGER – returns the mode used to trigger the measurement.*

  - MESSAGE
  - PRESSURE
  - BUTTON
  - BOTH

- *MEASMASK – returns the process mask used to create the measurement. The details of the bit settings are described under factory only PARAMETERS elsewhere in this document.*

- *index – number of the desired stored measurement. An index of zero selects the last measurement.*
- *geometry – required for any measurement data that also requires an index.*
  - *SPIN – Specular Included.*
  - *SPEX – Specular Excluded.*

Response

This command returns 31 floating point values if SPIN, SPEX, or REFL is specified. There is a value for each wavelength from 400nm to 700nm at 10nm increments. If UseExtendedRefls is enabled through Params, then 40 floating point values are returned instead, ranging from 360nm to 750nm.

Colorspace options return one to three floating point values respectively.

Other options return data as indicated in their respective descriptions above.

Examples

Fetch SPIN reflectance values for the last measurement

SAMPLE GET REFL 0 SPIN
90.3412 91.5613…(31 or 40 Spin values)

Fetch the SPIN LAB values from the fifth stored measurement

SAMPLE GET LAB 5 SPIN
92.6315 0.1356 3.2561

Fetch the SPEX LAB values from the fourth stored measurement

SAMPLE GET LAB 4 SPEX
91.6315 0.1256 3.1561

Fetch the temperature for the last measurement.

SAMPLE GET TEMPERATURE 0 SPIN
31.31

SerialNum Get

serialnum get arg

Retrieves the specified serial number.

Arguments

- *arg – the serial number to retrieve*
  - *INST – instrument serial number*
  - *OEM – original equipment manufacturers model number*
  - *MODEL – instrument model type*

Response

Returns the value of the specified serial number.

Examples

Get the instrument serial number

SERIALNUM GET INST
123456

Get the model

SERIALNUM GET MODEL
CI52

Standard Clear

standard clear option

Used to delete one or more standards stored in the instrument.

Arguments

- *option*
  - *ALL – Deletes all the standards in the current project.*
  - *A number indicating which stored standard is to be deleted. This number is relative to the current active project.*
  - *A name of a standard in the current project to be deleted.*

Response

Returns command/status only. No data.

Example

Clear all the stored standards in the current project.

STANDARD CLEAR ALL
<00>

Clear a standard by name from the current project.

STANDARD CLEAR SunshineYellow
<00>

Clear the 6th standard in the current project.

STANDARD CLEAR 6
<00>

Standard Clone

standard clone

Used to make the active sample into a new standard. The cloned standard will use reflectance from the current sample.

Arguments

None.

Response

Returns command/status only. No data.

Example

Clone the current standard.

STANDARD CLONE
<00>

Standard Edit

standard edit arg

Prepares the standard specified by *arg* for edits. To save any changes made after calling *edit* *standard store* and *project_store* must be called.

Arguments

- *arg*
  - *project relative standard num (1 through number of standards in project). Specifying 0 will edit the currently active standard*
  - *standard name*

Response

Returns command/status only. No data.

Example

Edit the first standard in this project

STANDARD EDIT 1
<00>

Edit the standard name Blue

STANDARD EDIT Blue
<00>

Standard Get

---

standard get option {index} {argument1} {argument2}

Used to retrieve a standard and tolerances from the instrument.

Arguments

- *option*
  - *NAME – Returns the name of the standard up to 32 ASCII-characters or UTF-16 characters.*
  - *GUID – Returns a Globally Unique IDentifier. This is a 36 character string of hexadecimal characters.*
  - *REFL – Returns the 31 reflectance values of the standard (requires the geometry, SPIN or SPEX, to be specified).*

    - *argument1* – The geometry of the reflectance data.

- **SPIN** – Specular included.
- **SPEX** – Specular excluded.

- *COUNT – Returns the number of standards stored in the instrument for the active project. No other arguments are required.*
- *AVAILABLE – Returns the number of standard openings are still available for the active project. It also takes into account the maximum number of total standards (all projects combined) allowed in the instrument.*
- *ACTIVE – Returns the number and the name of the active standard. If no standard is selected, this command will return the 0xA3 (STATUS_STANDARD_DOES_NOT_EXIST) status code.*
- *SIGNATURE – Returns the signature string which gets associated with each standard (same as a sample signature). It includes information pertaining to the conditions that existed when the standard measurement was made. The information is cryptic and involved and beyond the scope of this document. This command is intended only for use by specialized utilities that parse the data and act accordingly.*

  - *argument1* – if *argument1* for the signature is set to "parsed" the response to this command will be a parsed and formatted version of the standard signature.

- *TOLERANCE – Returns the tolerance data as specified in argument1.*

  - *argument1* – Tolerance option.
    - **COUNT** – Returns the number of tolerances stored with the standard. No other arguments are required.
    - **TYPE** – Returns the difference method for the tolerance.
      - "*DLAB*" – L*a*b* tolerance type.
      - "*DLCH*" – L*C*H* tolerance type.
      - "*DXYZ*" – XYZ tolerance type.
      - "*DYXY*" – Yxy tolerance type.
      - "*DLUV*" – L*u*v* tolerance type.
      - "*DHUNTERLAB*" – Hunter Lab tolerance type.
      - "*DELAB*" – Delta E tolerance type.
      - "*DECMC*" – Delta Ecmc tolerance type.
      - "*DE2000*" – Delta E 2000 tolerance type.
      - "*DE94*" – Delta E 94 tolerance type.
      - "*DYI98*" – E313-98 yellowness index tolerance type.
      - "*DYI73*" – E313-73 yellowness index tolerance type.
      - "*DYI25*" – D1925 yellowness index tolerance type.
      - "*DWI98*" – E313-98 whiteness index tolerance type.

- "*DWI73*" – E313-73 whiteness index tolerance type.
- "*DWIBERGER*" – Berger whiteness index tolerance type.
- "*DWIHUNTER*" – Hunter whiteness index tolerance type.
- "*DWISTENSBY*" – Stensby whiteness index tolerance type.
- "*DWITAUBE*" – Taube whiteness index tolerance type.
- "*SHADESORT*" or "*SHADESORTLAB*" – L*a*b* 555 Shade Sort tolerance type.
- "*SHADESORTLCH*" – L*C*H* 555 Shade Sort tolerance type.
- "*SHADESORTHUNTER*" – Hunter Lab 555 Shade Sort tolerance type.

- **GEOMETRY** – Returns the geometry of the tolerance.
  - "*SPIN*"
  - "*SPEX*"
- **HIGH** – Returns 1 to 3 high limit values of the tolerance.
- **LOW** – Returns 1 to 3 low limit values of the tolerance.
- **COEFF** – Returns 0 or more coefficient values if they are necessary for the tolerance.
- **ILLUMOBS** – Returns the illuminant/observer used by the tolerance.
  - "*A/2*"
  - "*A/10*"
  - "*C/2*"
  - "*C/10*"
  - "*D50/2*"
  - "*D50/10*"
  - "*D65/2*"
  - "*D65/10*"
  - "*F2/2*"
  - "*F2/10*"
  - "*F7/2*"
  - "*F7/10*"
  - "*F11/2*"
  - "*F11/10*"
  - "*F12/2*"
  - "*F12/10*"

- *argument2* – The number of the tolerance (1 – 6).

- *TAG1 – A 4-byte number containing information about the standard. Returned as hexadecimal value*

  - The first *nibble* holds the <u>UV settings</u> for this standard (0×0000000F)
  - The second *nibble* holds the <u>measlampmode</u> (0×000000F0)
  - The next *bit* holds whether or not this standard is locked from GUI edits (0×00001000)

    - The next 6 *bits* hold whether that tolerance id is a symmetric or asymmetric tolerance. (0×00007E00)

  - *TAG2 – The standard type:*

    - 0 = Not specified
    - 1 = Pantone
    - 2 = Downloaded

  - *Gloss – Get the standards loaded gloss value. The instrument does not calculate gloss for standards.*
  - *TimeStamp – Get the standards time stamp.*
- *index – The number of the standard (1 to the number of standards stored in current project). If 0 then the data for the active standard is retrieved. If the index is the word "NEW" then data residing in the new standard is retrieved (See also standard set for creating and loading a new standard).*

Response

STANDARD Status Data…

Example

Get the SPEX reflectance values from standard number 56.

STANDARD GET REFL 56 SPEX
90.3412 91.5613…(31 SPEX values)

Get the high limits from tolerance 3 of standard number 29.

STANDARD GET TOLERANCE 29 HIGH 3
90.00 1.50 2.99

Get the illuminant/observer from tolerance 10 of standard number 5.

STANDARD GET TOLERANCE 5 ILLUMOBS 10
D65_2

Get the geometry from tolerance 3 of standard number 101.

STANDARD GET TOLERANCE 101 GEOMETRY 3
SPIN

Get the name of the NEW standard that's being created (data for a new standard are loaded using the standard set command.

STANDARD GET NAME NEW
SunshineYellow

Get the SPIN reflectance values for the currently active standard.

STANDARD GET REFL 0 SPIN
91.2345 92.6543…(31 SPIN values)

Get the number and name of the active standard.

STANDARD GET ACTIVE
4 CottageWhite

Get the name of a UTF-16 named standard

STANDARD GET NAME 3
\u8124\u3234\u6754\u4354

Get the time stamp of a standard (default timestamp)

STANDARD GET TIMESTAMP 3
00:00:00 01/01/70

Get the time stamp of a standard

STANDARD GET TIMESTAMP 3
13:45:57 02/15/2011

Standard New

standard new name

Prepares the standard scratch space for the creation of a new standard. Loads the standard with defaults for reflectance and tolerance data.

- *Defaults*
  - GUID = *Blank*
  - REFL = *All reflectances are 0.0 for both SPIN and SPEX.*
  - TOLERANCE = *user defined defaults defined through the use of the default tolerance settings.*

Arguments

- *name*
  - *specifies the desired name for the new standard. Can be up to 32 ASCII or UTF-16 characters.*

Response

Returns command/status only. No data.

Example

Create a new standard

STANDARD NEW YellowGreen
<00>

Create a new standard

STANDARD NEW Yellow\u4568\u3423Green
<00>

Standard Select

standard select name

Used to select the active standard by its name or by using its project relative id number. If no name is provided then the number and name of the currently active standard is returned.

Arguments

- *name – The name given to the standard when it was created or the numeric index where the standard was stored in the database. The number and name of the current standard is returned if no name is provided.*

Response

Returns the number and name of the currently active standard if no name is provided. Otherwise, returns command/status only. No data.

Example

Select standard SunshineYellow to be the active standard.

STANDARD SELECT SUNSHINEYELLOW
<00>

Select the 23 standard in the active project.

STANDARD SELECT 23
<00>

Get the number and name of the currently active standard.

STANDARD SELECT
23 SunshineYellow

Standard Set

standard set option {argument1} {argument2} {data...}

Used to store a standard in the instrument. Each standard contains a name, a globally unique ID (GUID – used by the host software and ignored by the firmware), the reflectance data for each geometry (SPIN, SPEX), and a set of tolerances. There can be up to six tolerances specified, each with its own type, illum/obs, and geometry. Depending on the tolerance, there may also be some required high/low limits and/or coefficients.

The *STANDARD STORE* command is used to store the standard into the instrument's non-volatile database memory after all the data for the standard have been loaded.

Arguments

- *option – most of the options below also require data to be stored in the item unless indicated otherwise.*
  - *NAME – sets the name of the standard up to 32 characters (no spaces).*
  - *GUID – sets a Globally Unique IDentifier. This is a 36 character string of hexadecimal characters.*
  - *REFL – sets the 31 reflectance values of the standard (requires the geometry, SPIN or SPEX, to be specified).*

    - *argument1* – Specifies geometry of the reflectance data.
      - **SPIN** – Specular included.
      - **SPEX** – Specular excluded.
      - **SAMP** – Copies both SPIN and SPEX from the current sample.

- *SIGNATURE – This is string of coded data which includes information pertaining to the conditions that existed when the standard measurement was made. The information is cryptic and involved and beyond the scope of this document. The maximum signature length is 111 characters. A summary of information contained in the signature is as follows:*

  - Signature version
  - Instrument ID (20 characters maximum)
  - Cal plaque ID (16 characters maximum)
  - Wavelength interval
  - Minimum wavelength
  - Maximum wavelength
  - Measurement date
  - NetProfiler status
  - NetProfile creation date
  - NetProfile expiration date
  - NetProfile gloss value
  - Transform ID (16 characters maximum)
  - Transform gloss value

- *TOLERANCE – Sets the tolerance data as specified in argument1.*

  - *argument1* – Tolerance option.
    - **ENABLE** – Enables the tolerance (up to 6 tolerances may be enabled).
    - **DISABLE** – Disables the tolerance.
    - **ASYMMETRIC** – Used to set the tolerances as symmetric or asymmetric
      - Supply the tolerance number followed by a 1 to set that tolerance as asymmetric, 0 will make it symmetric
    - **TYPE** – Sets the difference method for the tolerance.
      - "*DLAB*" – Tolerance on the L*a*b* difference.
      - "*DLCH*" – Tolerance on the L*C*H* difference.
      - "*DXYZ*" – Tolerance on the XYZ difference.
      - "*DYXY*" – Tolerance on the Yxy difference.
      - "*DLUV*" – Tolerance on the L*u*v* difference.
      - "*DHUNTERLAB*" – Tolerance on the Hunter Lab difference.
      - "*DELAB*" – Tolerance on the delta E difference.
      - "*DECMC*" – Tolerance on the delta Ecmc difference.
      - "*DE2000*" – Tolerance on the delta E 2000 difference.
      - "*DE94*" – Tolerance on the delta E 94 difference.
      - "*DYI98*" – Tolerance on the E313-98 yellowness index difference.

- "*DYI73*" – Tolerance on the E313-73 yellowness index difference.
- "*DYI25*" – Tolerance on the D1925 yellowness index difference.
- "*DWI98*" – Tolerance on the E313-98 whiteness index difference.
- "*DWI73*" – Tolerance on the E313-73 whiteness index difference.
- "*DWIBERGER*" – Tolerance on the Berger whiteness index difference.
- "*DWIHUNTER*" – Tolerance on the Hunter whiteness index difference.
- "*DWISTENSBY*" – Tolerance on the Stensby whiteness index difference.
- "*DWITAUBE*" – Tolerance on the Taube whiteness index difference.
- "*SHADESORT*" or "*SHADESORTLAB*" – Tolerance on the L*a*b* 555 Shade Sort difference.
- "*SHADESORTLCH*" – Tolerance on the L*C*H* 555 Shade Sort difference.
- "*SHADESORTHUNTER*" – Tolerance on the Hunter Lab 555 Shade Sort difference.

- **GEOMETRY** – Sets the geometry of the tolerance.
  - "*SPIN*"
  - "*SPEX*"
- **HIGH** – Sets the high limit values of the tolerance. It is the maximum high deviation from the standard before it is considered failed. The number of values required is dependent on the tolerance type. However, three values are required. It is recommended unused values be set to zero. For SHADESORT these values will be in the range from 5 to 9.
- **LOW** – Sets the low limit values of the tolerance. It is the maximum low deviation from the standard before it is considered failed. The number of values required is dependent on the tolerance type. However, three values are required. It is recommended unused values be set to zero. For SHADESORT these values will be in the range from 1 to 5.
- **COEF** – Sets coefficient values of the tolerance. The number of values required is dependent on the tolerance type. However, five values are required. It is recommended unused values be set to zero.
  - dEcmc Coefficients:
    - Coef1 is the *L* coefficient (default = 2.0).
    - Coef2 is the *C* coefficient (default = 1.0).
    - Coef3 is not used.
    - Coef4 is not used.
    - Coef5 is not used.
  - dE2000 and dE94 Coefficients:
    - Coef1 is the *L* coefficient (default = 2.0).
    - Coef2 is the *C* coefficient (default = 1.0).
    - Coef3 is the *h* coefficient (default = 1.0).
    - Coef4 is not used.
    - Coef5 is not used.
  - Shade Sort Coefficients:
    - Coef1 is the size for L or L* (0.01 to 9.99).

        - Coef2 is the size for a, a*, or C* (0.01 to 9.99).
        - Coef3 is the size for b, b*, or H* (0.01 to 9.99).
        - Coef4 is not used.
        - Coef5 is not used.
    - **ILLUMOBS** – Sets the illuminant/observer used by the tolerance.
        - *"A/2"*
        - *"A/10"*
        - *"C/2"*
        - *"C/10"*
        - *"D50/2"*
        - *"D50/10"*
        - *"D65/2"*
        - *"D65/10"*
        - *"F2/2"*
        - *"F2/10"*
        - *"F7/2"*
        - *"F7/10"*
        - *"F11/2"*
        - *"F11/10"*
        - *"F12/2"*
        - *"F12/10"*
    - *argument2* – The number of the tolerance (1 – 6).

- ***TAG1*** *– Bitmask that contains certain properties for this standard*

    - The first *nibble* holds the <u>UV settings</u> for this standard (0×0000000F)
    - The second *nibble* holds the <u>measlampmode</u> (0×000000F0)
    - The next *bit* holds whether or not this standard is locked from GUI edits (0×00001000)
    - The next 6 *bits* hold whether that tolerance id is a symmetric or asymmetric tolerance. (0×00007E00)

- *TAG2* ***– The standard type:***

    - 0 = Not specified
    - 1 = Pantone
    - 2 = Downloaded

- *LOCK – Used to lock the standard from being edited on the GUI*

    - Set to 1 to lock this standard, 0 to unlock this standard

- *GLOSS – Used to set the standards Gloss Value*
- ***TimeStamp** – Used to set a standards creation time stamp*
  - argument is 32 bit number since Unix Epoch

Response

STANDARD Status

Example

Set the SPEX reflectance values.

STANDARD SET REFL SPEX 90.3412 91.5613…(31 SPEX values)
<00>

Set the high limits for tolerance 3 where the first value is allowed to rise by 2.00, the second 1.5, and the third by 1.0.

STANDARD SET TOLERANCE HIGH 3 2.00 1.50 1.00
<00>

Set the low limits for tolerance 3 where the first value is allowed to drop by 2.00, the second 1.5, and the third by 1.0.

STANDARD SET TOLERANCE LOW 3 -2.00 -1.50 -1.00
<00>

Set the illuminant/observer for tolerance 1.

STANDARD SET TOLERANCE ILLUMOBS 1 D65_2
<00>

Set the geometry for tolerance 6.

STANDARD SET TOLERANCE GEOMETRY 6 SPIN
<00>

Enable tolerance 1.

STANDARD SET TOLERANCE ENABLE 1
<00>

Disable tolerance 2.

STANDARD SET TOLERANCE DISABLE 2
<00>

Standard Store

standard store

Used to store the standard data that was previously loaded using the *standard set* commands. If the standard name is blank when this command is used, the name will be set to “Standard #”, where # is the index of the standard.

Arguments

Response

Returns command/status only. No data.

Example

Store the standard.

STANDARD STORE
<00>

System Query

system query {trait}

Requests information about this instrument for use by generic software applications.

Arguments

- *trait – The following traits are available:*
  - *aperture – This will return the presently active aperture size:*
    - SMALL
    - MEDIUM
    - LARGE
  - *geometries – This will return the geometries. For most models this will be SPIN SPEX*

- *wavelengths – This will return the wavelengths supported by this device. Usually this is 400 to 700 at 10nm intervals*

Response

- *Returns the status followed by the details listed above.*

Examples

Request the geometries for this device

SYSTEM QUERY geometries
SPIN SPEX

Transform Get

---

transform option

Used to get the active transform name or a list of available transforms.

Arguments

- *option – This is the item to get.*
  - *name – Returns the name of the active transform.*
  - *names – Returns the names of all the available Transforms.*

Response

Returns command/status and the data specific to the option selected.

Example

Get the Transform name

TRANSFORM GET NAME
SOMETRANSFORM

Get a list of available transforms

TRANSFORM GET NAMES
SOMETRANSFORM ANOTHERTRANSFORM

Transform Select

transform select name

Used to select which Model Transform is active. The name “Off” is reserved to turn off Transforms.

Arguments

- *name – The desired text string name for the Transform. The name “Off” disables all Transforms.*

Response

Returns command/status only. No data.

Example

Select the Transform named ‘SomeTransform’

TRANSFORM SELECT SOMETRANSFORM
<00>

Turn off Transforms

TRANSFORM SELECT Off
<00>

Version Get

version get

Returns the firmware model, version, and datecode for the running application / monitor.

Arguments

none

Response

Returns the version information of the instrument. The information is returned as text, in the following order:

- *Instrument Model Number.*
- *The Date Code when this version of firmware began production.*

- *The version of the firmware on the instrument.*

Examples

Request the instrument version information

VERSION GET
X-Rite CI52 09B14 1.0.54

USB Error Codes

Error Code	Error Title	Description
(0xE1000000)	USB_DEVICE_NOT_FOUND	USB Device not found
(0xE1000001)	USB_FAILED_OPEN	Unable to open the USB device for reading
(0xE1000002)	USB_FAILED_OPEN_WRITE	Unable to open the USB device for writing
(0xE1000003)	USB_SEND_NOT_CONNECTED	Unable to transmit data because the device is not connected
(0xE1000004)	USB_SEND_TIMEOUT	Data not transmitted within the timeout period
(0xE1000005)	USB_RECEIVE_NOT_CONNECTED	Unable to receive data because the device is not connected
(0xE1000006)	USB_RECEIVE_TIMEOUT	Data not received within the timeout period
(0xE1000007)	USB_RECEIVE_PACKET_LENGTH	BCP packet length is too short
(0xE1000008)	USB_RECEIVE_BAD_BCP	Invalid BCP packet
(0xE0001021)	USB_DEVICE_OPENED	USB device has been opened by a different process
(0xE0001000)	USB_FAILED	USB operation failed
(0xE0001001)	USB_INVALID_INBUFFER	Invalid USB driver input buffer
(0xE0001002)	USB_INVALID_OUTBUFFER	Invalid USB driver output buffer
(0xE0001003)	USB_OUT_OF_MEMORY	USB Driver out of memory
(0xE0001004)	USB_PENDING_REQUESTS	USB read or write requests are pending
(0xE0001006)	USB_NOT_CONFIGURED	USB device is not configured
(0xE0001007)	USB_OPEN_PIPES	There are open USB pipes
(0xE0001009)	USB_NOT_BOUND	USB handle is not bound to a pipe
(0xE000100A)	USB_DEVICE_NOT_PRESENT	USB device is not present
(0xE000100B)	USB_CONTROL_NOT_SUPPORTED	USB control code is not supported
(0xE000100C)	USB_TIMEOUT	USB operation has timed out
(0xE000100D)	USB_INVALID_RECIPIENT	Invalid USB recipient
(0xE000100E)	USB_INVALID_TYPE	Invalid USB request
(0xE000100F)	USB_INVALID_IOCTL	Invalid USB IOCTL code
(0xE0001010)	USB_INVALID_DIRECTION	Invalid USB pipe direction
(0xE0001012)	USB_POOL_EMPTY	USB driver is out of memory
(0xE0001013)	USB_PIPE_NOT_FOUND	USB pipe not found
(0xE0001015)	USB_OUT_OF_ADDRESS_SPACE	USB driver is out of resources
(0xE0001016)	USB_INTERFACE_NOT_FOUND	USB interface not found
(0xE0001017)	USB_INVALID_DEVICE_STATE	Invalid USB device state
(0xE0001018)	USB_INVALID_PARAM	Invalid USB IOCTL parameter

(0xE000101A)	USB_INVALID_POWER_STATE	Invalid USB power state
(0xE000101B)	USB_POWER_DOWN	USB device has powered down
(0xE000101D)	USB_SET_CONFIGURATION_FAILED	USB set configuration failed
(0xE000101F)	USB_INVALID_PROCESS	USB operation not allowed for this process
(0xE0001020)	USB_DEVICE_ACQUIRED	USB device has been acquired by a different process
(0xE1000009)	USB_HOST_CONTROLLER_ERROR	USB Host Controller Error
(0xE100000A)	USB_BUS_DRIVER_ERROR	USB Bus Driver Error

Instrument Command Status Codes

Error Code	Error Title	Description
0×00	STATUS_OK	No error
0×01	STATUS_BAD_CMD	Unrecognized command
0×02	STATUS_BAD_CMD_ARG	Invalid value for a command argument
0×03	STATUS_BAD_CMD_FORMAT	Missing a necessary argument or contains an extra argument
0×04	STATUS_OP_TIMEOUT	Operation took too long
0×05	STATUS_OP_PENDING	Operation is still processing
0×06	STATUS_OP_WRONG_MODE	Operation not valid for current mode
0×07	STATUS_OP_NONE_YET	Operation has not been performed yet
0×08	STATUS_OP_WRONG_OPERATION	Operation requested was not the one performed
0×09	STATUS_OP_INVALID_OPERATION	Operation requested was not valid
0x0A	STATUS_OP_SECURITY_FAIL	Feature not available due to lack of authorization
0×0B	STATUS_SCP_PACKET_TOO_BIG	The packet is too large for the instrument to handle
0×0C	STATUS_SCP_PACKET_TOO_SMALL	The packet is less than the size of an empty frame
0×0D	STATUS_SCP_INVALID_START_BYTE	The start byte was incorrect for the packet sent
0×0E	STATUS_SCP_INVALID_STOP_BYTE	The stop byte was incorrect for the packet sent
0×0F	STATUS_SCP_INVALID_PAYLOAD_LEN	The payload length specified doesn’t match the real length
0×18	STATUS_PARAM_BAD_NAME	Invalid Parameter Specified
0×1A	STATUS_PARAM_BAD_RANGE	Invalid Parameter Value Specified
0x3D	STATUS_NO_EXT_POWER	Procedure requires and external power supply

		to be connected
0×3E	STATUS_CLOCK_ERROR	Invalid Date Specified
0×6A	STATUS_TEMP_OUT_OF_RANGE	The Instrument Temperature is out of range
0×6B	STATUS_TRIGGER_RELEASED	The trigger was released during a measurement
0×6C	STATUS_TEMP_DELTA_OUT_OF_RANGE	The instruments measurement temperature is out of range from the calibration temperature
0×6D	STATUS_APERTURE_CHANGED	The instrument has detected the aperture switch has changed positions
0×6F	STATUS_TEMP_UV_DELTA_OUT_OF_RANGE	The instruments measurement temperature is out of range from the calibration temperature
0×81	STATUS_CAL_BAD_TEMPERATURE	Temperature changed too much during the calibration process
0×83	STATUS_CAL_BAD_LAMP	Instrument light source is weak or burned out
0×84	STATUS_CAL_NEEDS_BLACK	No Black Calibration has been performed
0×85	STATUS_CAL_NEEDS_WHITE	No White Calibration has been performed
0×8B	STATUS_CAL_PLAQUE_WRONG_BLACK	You're Calibrating on something other than the black trap
0×8C	STATUS_CAL_PLAQUE_WRONG_WHITE	You're Calibrating on something other than the white plaque
0×8D	STATUS_CAL_PLAQUE_DIRTY	The white plaque looks different
0×8E	STATUS_CAL_EXPIRED	Your calibration has expired
0xA0	STATUS_MAX_STORAGE_REACHED	Memory space depleted or a storage limit is reached
0xA2	STATUS_SAMPLE_DOES_NOT_EXIST	The selected sample was not stored
0xA3	STATUS_STANDARD_DOES_NOT_EXIST	The selected standard was not stored
0xA4	STATUS_TOLERANCE_DOES_NOT_EXIST	The selected tolerance was not stored
0xA5	STATUS_NO_STANDARD_ACTIVE	A comparison was attempted without choosing a standard
0xAB	STATUS_PROJ_ALREADY_EXISTS	Duplicate project names are not allowed
0xAC	STATUS_PROJ_STDS_FULL	The project standard limit is reached
0xAD	STATUS_PROJ_MAX_PROJ	The project limit is reached
0xAE	STATUS_STANDARD_CANT_EDIT	Locked standards cannot be changed
0×E4	STATUS_SERNUM_NOT_SET	The OEM serial number is not set

Parameters

Parameter	Default	Min	Max	Desc
AllowProjectSelect	1	0	1	Allow the selection of a project thru GUI when 1
AllowSampleDelete	0	0	1	Allow deleting of samples thru GUI when 1
AllowSampleHistory	1	0	1	Allow navigation of sample history thru GUI when 1
AllowSampleStore	0	0	1	Allow samples to be stored when 1
AllowStdAuto	1	0	1	Allow standards to be automatically chosen when 1
AllowStdCreate	0	0	1	Allow standards to be created from samples when 1
AllowStdDelete	0	0	1	Allow deleting of standards thru GUI when 1
AllowStdEdit	1	0	1	Allow standards to be edited thru GUI when 1
AutoNameSample	0	0	2	Selects the AutoNameSample prefix format for samples
AutoNameStandard	0	0	1	Selects the AutoNameStandard prefix format for standards
ColorSpace	1	0	19	Selects the current ColorSpace
ColorTable	0	0	1	Selects the current ColorTable to be used by the instrument
DateFormat	0	0	0	Selects the current DateFormat
Geometry	0	0	1	Selects the current Geometry to show data for
Handedness	0	0	1	Selects the current Handedness of the instrument
IllumObs	9	0	23	Selects the current IllumObs
Language	0	0	9	Selects the current Language
MeasAverageNum	1	1	99	Selects the number of readings are averaged to create a measurement
MeasLampMode	0	0	1	Selects the current LampMode
Metamerism1	0	0	15	First IllumObs selection for metamerism
Metamerism2	0	0	15	Second IllumObs selection for metamerism
OpacityMode	0	0	2	The OpacityMode of the instrument
PowerManagement	2	0	3	Selects the current PowerManagementMode
ProfileAccessMask	0xFFFEFDFF	0	0xFFFFFFFF	The Access Mask (low bits) for options to show in the config screen
ProfileAccessMask2	0×0000000F	0	0xFFFFFFFF	The Access Mask (high bits) for options to show in the config screen
ProjConfigMask	0xFFFFFFFF	0	0xFFFFFFFF	Determines the global and project specific features

				allowed for editing projects
ShadeSortEnable	0	0	1	1 allows shade sort output to be displayed in the delta measure screens
ShowGraph	0	0	1	Show the graph version of colorspace if supported on gui when 1
StrengthMode	0	0	2	Selects the current StrengthMode
TolWarnThreshold	80	0	100	Selects the percentage of a limit needed to trigger a warning for the Pass/Fail feature
VerbalDiff	0	0	1	1 allows differences between sample and standard to be displayed verbally instead of numerically
UVSetting	0	0	2	Select which UV Setting should be used
WarningOverride	3	0	0xFFFFFFFF	Enables/Disables certain Warnings
CalStatusHappyColor	0	0	10	Change the default color of the calibration LED when a valid calibration. 0 = Green, not 0 = Yellow
CalWhiteTimeout	24	0	0xFFFFFFFF	Number of hours that white cal is valid
MeasTriggerMode	1	0	3	The method to trigger a measurement
UseExtendedRefls	0	0	1	Outputs 40 Reflectance points instead of 31 when set to 1 (extra are just extended endpoints)
VolumeLevel	2	0	2	The volume level for the beeper (0 = off, 1 = low, 2 = high)

Color Spaces

0	Pass / Fail Summary
1	Lab
2	LCh
3	XYZ
4	Yxy
5	Luv
6	Hunter Lab
7	Reflectance
8	Munsell Notation
9	Gray Scale
10	Yellowness Index E313-98

11	Yellowness Index E313-73
12	Yellowness Index D1925
13	Whiteness Index E313-98
14	Whiteness Index E313-73
14	Whiteness Index Berger
16	Whiteness Index Hunter
17	Whiteness Index Stensby
18	Whiteness Index Taube
19	Metamerism
20	Metamerism D6172
21	Gloss
22	Delta Lab
23	Delta LCH
24	Delta E(CIELab)
25	Delta Ecmc
26	Delta E 2000
27	Delta E 1994
28	Delta XYZ
29	Delta Yxy
30	Delta Luv
31	Delta Hunter Lab
32	Delta Reflectance
33	Delta Yellowness Index E313-98
34	Delta Yellowness Index E313-73
35	Delta Yellowness Index D1925
36	Delta Whiteness Index E313-98
37	Delta Whiteness Index E313-73
38	Delta Whiteness Index Berger
39	Delta Whiteness Index Hunter
40	Delta Whiteness Index Stensby
41	Delta Whiteness Index Taube

Date Formats

0	MM/DD/YYYY
1	DD/MM/YYYY
2	DD.MM.YYYY
3	YYYY/MM/DD
4	YYYY-MM-DD

Illum Obs

0	A/2
1	A/10
2	C/2
3	C/10
4	D50/2
5	D50/10
8	D65/2
9	D65/10
12	F2/2
13	F2/10
14	F7/2
15	F7/10
20	F11/2
21	F11/10
22	F12/2
23	F12/10

Strength Modes

0	Apparent
1	Chromatic
2	Tristimulus

Languages

0	English
1	Chinese Simplified
2	Chinese Traditional
3	German
4	Spanish
5	French
6	Italian
7	Japanese
8	Korean
9	Portugese

Geometries

0	SPIN
1	SPEX

Handedness

0	RIGHT HANDED
1	LEFT HANDED

Sample Auto Name Settings

0	No auto name prefix (Name is left blank)
1	Prefix + text. *Example: Sample -*
2	Prefix + sample number. *Example: Sample – 23*

Standard Auto Name Settings

0	No auto name prefix (Name is a time stamp)
1	Prefix + standard number. *Example: Standard5*

Color Table

0	Color Table 5
!0	Color Table 6

Opacity Modes

0	Over Black
1	Over White
2	Color at 100%

Lamp Modes

0	Tungsten Only
1	UV included

UV Setting

0	D65/10
1	Adjust 1
2	Adjust 2

Warning Override

Bit	Mask	Purpose
0	0×00000001	Enables the lamp intensity warning during calibrations
1	0×00000002	Enables the aperture changed warning on instruments with switchable apertures

Power Management Modes

0	None
1	Min
2	Normal
3	Max

Profile Access Mask (low bits)

Bit	Mask	Purpose
0	0×00000001	Trigger Mode
1	0×00000002	Measurement Averaging Mode
2	0×00000004	Calibration Timeout
3	0×00000008	Sample Storage
4	0×00000010	Sample Delete
5	0×00000020	Sample History
6	0×00000040	Auto Standard
7	0×00000080	Standard Creation
8	0×00000100	Standard Deletion
9	0×00000200	Project Select
10	0×00000400	Volume Level
11	0×00000800	Power Management
12	0×00001000	Language Select
13	0×00002000	Date Format
14	0×00004000	Clear All Samples
15	0×00008000	Load Defaults
16	0×00010000	Profile
17	0×00020000	Color Table
18	0×00040000	Net Profiler
19	0×00080000	Transform

20	0×00100000	Strength
21	0×00200000	Metamerism 1
22	0×00400000	Metamerism 2
23	0×00800000	Edit Tolerances
24	0×01000000	Edit Date and Time
25	0×02000000	Colorspaces
26	0×04000000	Illuminant/Observers
27	0×08000000	Opacity
28	0×10000000	Shade Sort
29	0×20000000	Verbal Difference
30	0×40000000	Whiteness Timeout
31	0×80000000	UV Whiteness Cal

Profile Access Mask (high bits)

Bit	Mask	Purpose
0	0×00000001	Measure Type
1	0×00000002	Handedness
2	0×00000004	Indices
3	0×00000008	UV Whiteness Number
4	0×00000010	Not Used
5	0×00000020	Not Used
6	0×00000040	Not Used
7	0×00000080	Not Used
8	0×00000100	Not Used
9	0×00000200	Not Used
10	0×00000400	Not Used
11	0×00000800	Not Used
12	0×00001000	Not Used

13	0×00002000	Not Used
14	0×00004000	Not Used
15	0×00008000	Not Used
16	0×00010000	Not Used
17	0×00020000	Not Used
18	0×00040000	Not Used
19	0×00080000	Not Used
20	0×00100000	Not Used
21	0×00200000	Not Used
22	0×00400000	Not Used
23	0×00800000	Not Used
24	0×01000000	Not Used
25	0×02000000	Not Used
26	0×04000000	Not Used
27	0×08000000	Not Used
28	0×10000000	Not Used
29	0×20000000	Not Used
30	0×40000000	Not Used
31	0×80000000	Not Used

Handheld Measure Trigger Modes

0	Command Only
1	Command or Shoe
2	Command or Button
3	Command or Shoe AND Button

Project Config Mask

Bit	Mask	Name	Meaning
0	0×00000001	Add Project	Allows users to add projects through the GUI. Applies globally on instrument
1	0×00000002	Edit Project	Allows users to edit projects through the GUI. Set on a per project basis
2	0×00000004	Delete Project	Allows users to delete projects through GUI. Set on a per project basis
4	0×00000010	Edit Project Name	Allows users to change the project name through GUI. Set on a per project basis
5	0×00000020	Add Standards	Allows users to add standards to the project through the GUI. Set on a per project basis
6	0×00000040	Delete Standards	Allows users to delete standards from the project through the GUI. Set on a per project basis
28	0×10000000	Unlock-able	Project can be unlocked. Set on a per project basis
31	0×80000000	Project	